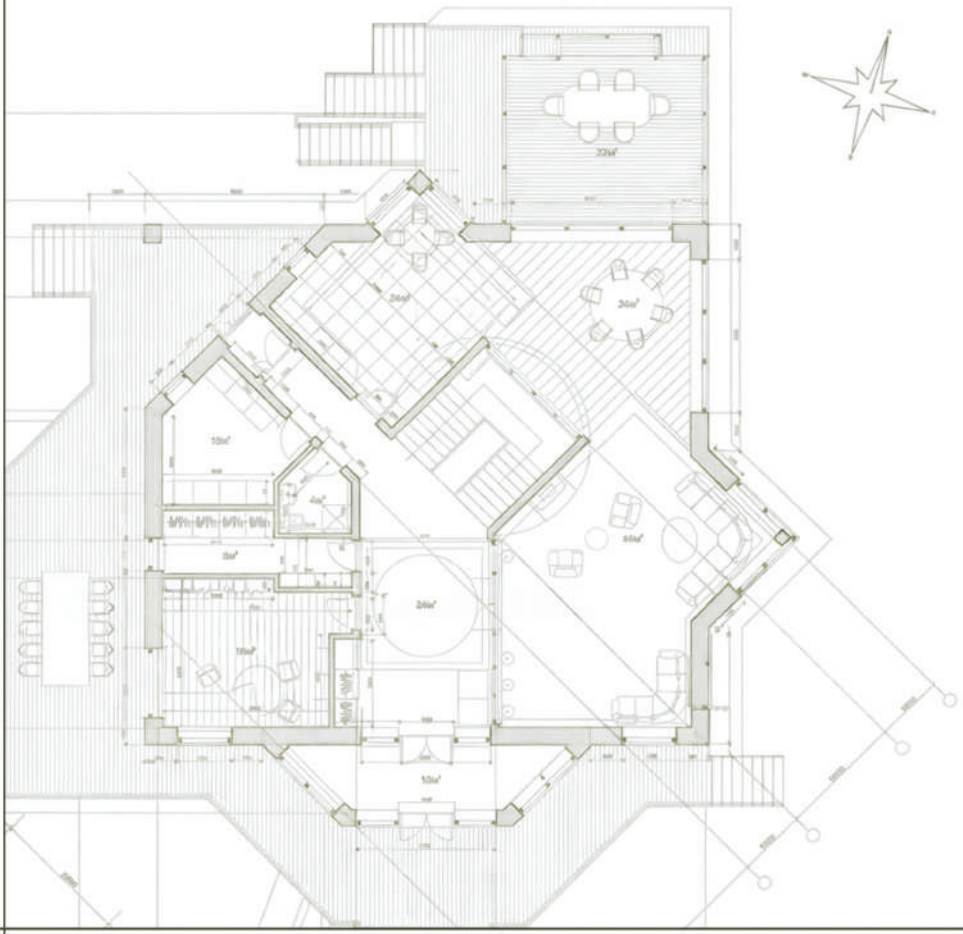
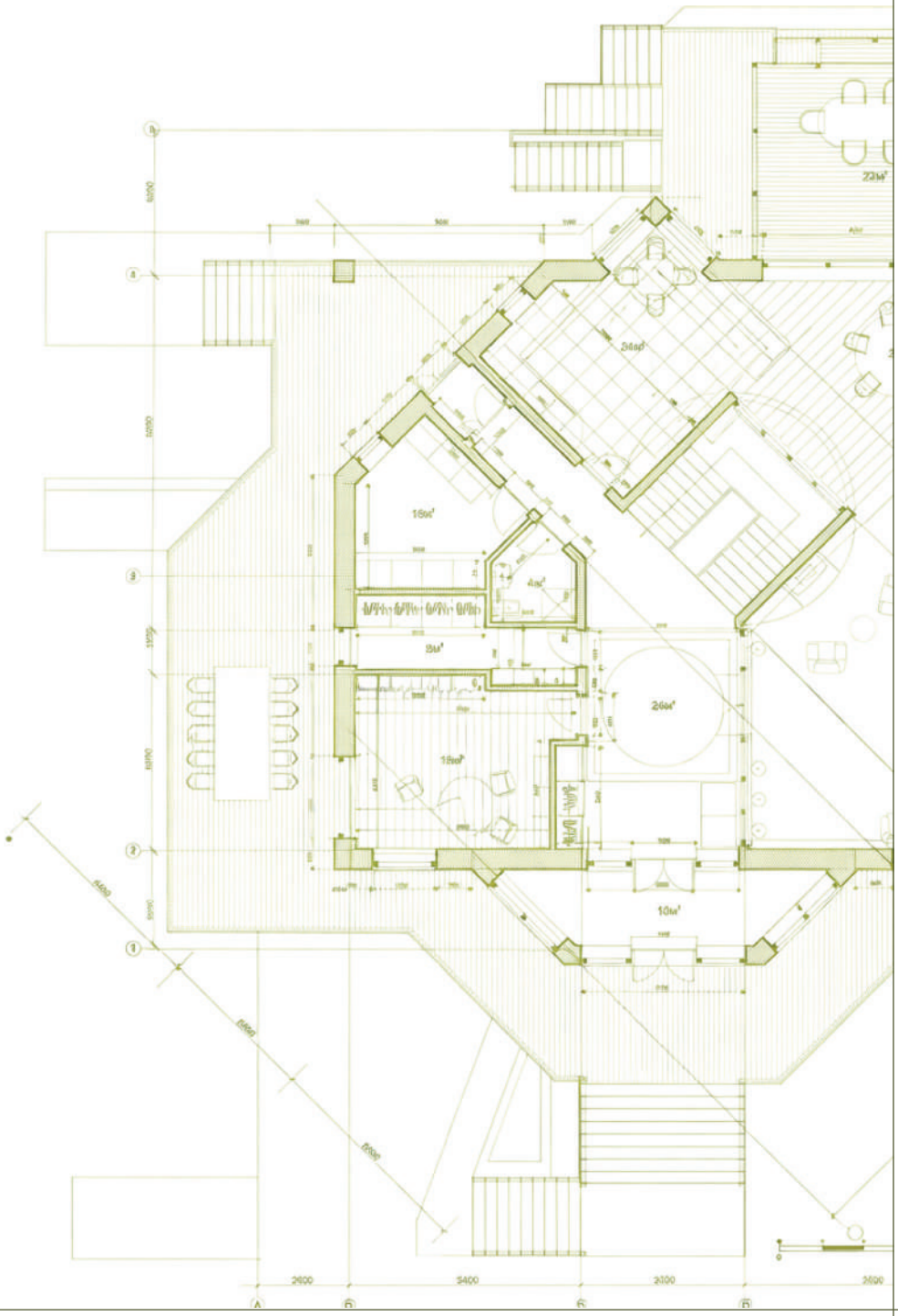
STONELAM®

THE FINEST NATURAL FACADES

CERTIFICATIONS AND TEST REPORTS

AT STONELAM, PERFORMANCE IS NEVER A CLAIM.
IT IS DOCUMENTED, TESTED, AND CERTIFIED.
EVERY SURFACE IS BACKED BY RIGOROUS TESTING
AND **GLOBALLY RECOGNIZED CERTIFICATIONS.**




23м²
34м²
16м²
4м²
9м²
12м²
26м²
10м²
2600
5400
2100
2600
А
Б
Б
Б

Certificates :-

25+ Years Warranty	01
Fire Resistant	07
Impact Load Test	09
Frost Resistant	16
Water Absorption	19
Wind Load Test	22
Adhesion Test	42
Food Grade Test	47

25+ Years Warranty Certificate

01



LAMINAM
SUPERIOR NATURAL SURFACES

Certificate

Façade application | WARRANTY

25
YEARS

We are architects of our environment, we pursue uniqueness.

LAMINAM
SUPERIOR NATURAL SURFACES

CONVENTIONAL WARRANTY FOR LAMINAM PRODUCTS USED FOR FAÇADES

This conventional Warranty applies to LAMINAM products when used for the construction of building façades. The Warranty is provided by the manufacturer, LAMINAM S.p.A., whose slabs are produced in the following three factories:

- **LAMINAM FIORANO:** Via Ghiarola New, 258 | 41042, Fiorano Modenese (MO) | Tel. +39.0536.1844200 - Fax +39.0536.1844201 | info@laminam.com
- **LAMINAM BORGO VAL DI TARO:** Via Primo Brindani 1 | 43043, Borgo Val di Taro (PR) | Italy | Tel. +39 0525 97864
- **LAMINAM LLC:** Kaluzhskaya oblast | Borovskiy raion | Dobrino village 10 Vostochniy proezd building1 | Russian Federation | Tel. +7 495 274 03 00

to which all eligible parties can address any requests as per the conditions described below.

Art. 1
This conventional warranty is offered by LAMINAM S.p.A. exclusively in favour of the direct Purchaser of the LAMINAM product, by which is meant the Customers of LAMINAM S.p.A. who receive the supply of the LAMINAM product from the latter and use it in the construction of building façades, regardless of the application system.
The application of this warranty, which is in addition to the ordinary guarantee due by the seller under the provisions of Italian law, is expressly excluded in the event that the same product is received by the user, for any reason, from parties other than the manufacturer LAMINAM S.p.A. or in cases where the product is used for purposes other than those expressly and unconditionally indicated in the previous paragraph.
This Warranty is also provided for LAMINAM surfaces with "Filo" Series finish and with finishes having a glossy surface produced by an after-firing processing of the slab (Lucidato Collection series type), regardless of their intended use, limited to the characteristics set forth in points b), c), d) of the following Art. 4.

Art. 2
Pursuant to this Warranty, the Manufacturer undertakes towards the Purchaser, as exactly identified in art. 1 above, who has purchased the new product and has used it for the construction of building façades, independently of the application system, under the mandatory conditions provided for in this document, to carry out the only replacement free of charge - with a product equal or similar in technical specifications and dimensions as better specified in art. 8 below - of the product recognised as affected by the faults and defects in relation to the specific features identified in art. 4 below, within an appropriate deadline. This deadline will be determined by the Manufacturer in relation to the extent of the specifications of the product to be replaced, excluding any other ancillary activity - among others processing and installation as described in art. 5 below, and other charge, compensation and/or indemnity for both consequential damage and loss of profit, even if the same are due to product defect. In particular, all charges necessary for the removal of the damaged slabs, the installation of the new slabs supplied in replacement of the first ones and the consequent restoration works and, in general, any other compensation and/or indemnity for direct, indirect and consequential damage, expenses and loss of profit, even if deriving from the impossibility of using the building or from interruptions to the activities carried out in the building where the product was used for the façade(s), are not included.

Art. 3
LAMINAM S.p.A. guarantees to the Purchaser the quality of its LAMINAM product when used in the construction of building façades and regardless of the application system used, for the duration of 25 years, starting from the date of purchase, which must be proved by the eligible person exclusively by means of the document for tax purposes (invoice), issued by the Manufacturer at the time of sale.

Art. 4
The product bearing the LAMINAM brand consists of solid porcelain stoneware slabs with any reinforcing fibres applied to the back, with maximum dimensions of 1000x3000mm, 1200x3000mm and 1620x3240mm, whose physical and mechanical specifications are indicated in the technical data sheet accompanying the product.
Due to its high level of resistance to chemical and physical stress, to the action of weather agents and to the non-absorbency of the product, LAMINAM S.p.A. guarantees, for the duration specified in article 2 above:

a) the inalterability of the colours and the surface after prolonged exposure to light (fastness of colour to light DIN 51094: no surface alteration after testing);
b) the impermeability of the surface (ISO 10545-3: average water absorption ≤ 0.3%);
c) the frost resistance (ISO 10545-12: no slab damaged after 100 cycles from -5°C to +5°C; ASTM C1026: no damage after 15 cycles);
d) resistance to thermal shock (ISO 10545-9: no slab damaged after 10 cycles from 145°C to 15°C; ASTM C484: no sample damaged);
e) scratch hardness of surface in case of contact with materials softer than ceramics, for example metals such as iron and aluminium (UNI EN 101: Mohs scale value ≤ 6);
f) chemical resistance (ISO 10545-13: A to B, according to the classes listed in the catalogue);
g) resistance to permanent stains (ISO 10545-14: according to the classes listed in the catalogue).

Due to the technical specifications and peculiarities of the product, cleaning of the slabs will be particularly easy and effective. However, any claims for damage to the slabs resulting from cleaning operations that are not covered by the Cleaning and Maintenance Manual, to which reference should be made, or that are carried out in a manner inconsistent with the provisions of the same Manual, will not be covered by this Warranty and will therefore be rejected, being the Manual regarded as an integral and substantial part of the Warranty.

LAMINAM S.p.A. - Società iscritta al registro delle imprese di Modena al n. 01969990355 | Sede Legale e Operativa: Via Ghiarola Nuova, 258 - 41042 Fiorano Modenese (MO) Italy | T. +39 0536 1844200 | cap. soc. € 7.200.000 R.E.A. 332793 | C.F. e P.IVA IT01969990355 SDI: MZO2AØU.

Art. 5
The Manufacturer shall only be liable for defects relating to the quality of the product, i.e. defects of origin and/or structural defects, attributable to the processing of the slabs, whereas it shall not be liable for any defects of the product resulting from its transformation, processing and installation.
This Warranty does not cover defects which are due to a cosmetic textural inconsistency and all defects arising from subsequent stages of production, such as:
- transport and handling in general;
- processing, which is to be understood as any activity of shaping, drilling and cutting of the slab, also for the implementation of the anchoring systems chosen by the installer, shaping of the corners, as well as polishing of the edge;
- installation, to be understood as any activity necessary for its installation and cutting;
- application and/or coupling with other materials subject to dimensional and/or endurance variations, also due to weather agents, humidity, environmental conditions.

Given that the special specifications of resistance to chemical-physical stresses concern the undamaged slabs, correctly stored and preserved, this Warranty is provided by LAMINAM S.p.A. only for the slabs that have not been subjected to processing and installation work, as indicated above, and any conventional warranty for the side surfaces, edges and corners, even if resulting from processing and installation work, remains expressly excluded.
In any case, this Warranty does not cover cases of improper and/or inadequate use and mistreatment of the product, even if attributable to the Purchaser's assignees and in any case to the final user of the product, such as for example:
a) improper or inadequate use, care or maintenance of the product and, in general, non-compliance with the indications and specifications contained in the Cleaning and Maintenance manual accompanying the product;
b) colour differences between the sample shown at the time of purchase and the product delivered;
c) physical, chemical and mechanical misuse;
d) improper preparation or maintenance of the installation place;
e) use as flooring;
f) use of the product in abnormal conditions and/or in conditions that lead to exceeding the tolerance and resistance limits indicated in the previous art. 4, letters a), b), c), d), e), f) and g) or in any case not in function of the technical specifications for which it was designed;
g) accidental damage and/or due to unforeseeable circumstances and/or force majeure;
h) damage caused, directly and/or indirectly, to people, things and animals as a result of behaviour falling within the hypotheses of express exclusion from the operation of the warranty, as indicated in points a), c), d), e), f) and g) above, or from failure to comply with the specifications indicated in the Maintenance and Cleaning Manual accompanying the product.
i) Breakage/crazing due to or caused by the incorrect dimensioning of the structures supporting the slabs, and/or events directly or indirectly attributable to the construction of the substructure, frames, fastening systems and installation.

Art. 6
To make use of the rights envisaged in this Warranty, it is the responsibility of the Purchaser of the LAMINAM product used for the construction of façades to notify LAMINAM S.p.A., at the a.m. addresses, of the defect found on the product by registered letter with return receipt within and no later than fifteen days from the discovery of the defect. The claim must be accompanied by a copy of the sales invoice issued by LAMINAM S.p.A., through which the purchase date and, if different, the delivery date, as well as the identification details of the product, will be proved. In order for this Warranty to be effective, it is therefore necessary that the above documentation is duly kept, sent in copy with the claim of defects and shown in original to the personnel appointed by LAMINAM S.p.A. in case of intervention.

Art. 7
This Warranty is valid, limited to the events and cases expressly provided for above, only if the Buyer, its assignees and in any case the end user, have complied with the prescriptions indicated in the Cleaning and Maintenance manual accompanying the product, and only if the Buyer has contacted LAMINAM S.p.A. to enforce this Warranty.

Art. 8
During the period of validity of the Warranty, the replacement supply will be carried out entirely free of charge by the personnel appointed by LAMINAM S.p.A. Since raw materials present in nature (clays, feldspars) are used for the processing of these products and special collections are created for them, according to technical and aesthetic choices that are continuously improved, LAMINAM S.p.A. undertakes to replace the product, if necessary and falling within the cases provided for by this Warranty, for the whole duration of the Warranty itself, with another product, at its own exclusive judgement, similar - in type, thickness, dimensions, technical specifications and colour - to the one to be replaced, chosen from those in production and available at the time when the need to proceed with the replacement arises. LAMINAM S.p.A., therefore, does not guarantee perfect uniformity between the replaced product and the one originally supplied and cannot in any way be held responsible for any difference in colour, type, thickness, dimensions and technical features, nor for the overall aesthetic appearance of the installation.

Art. 9
This Warranty may not be transferred, as it is conventionally provided by the Manufacturer to the Purchaser, as precisely stated in article 1 above.

Art. 10
This Warranty is governed exclusively by Italian law. Any dispute relating to the interpretation, execution and operation in general shall be submitted exclusively to the jurisdiction of the Italian courts and to the competence of the Court of Modena.

LAMINAM S.p.A. - Società iscritta al registro delle imprese di Modena al n. 01969990355 | Sede Legale e Operativa: Via Ghiarola Nuova, 258 - 41042 Fiorano Modenese (MO) Italy | T. +39 0536 1844200 | cap. soc. € 7.200.000 R.E.A. 332793 | C.F. e P.IVA IT01969990355 SDI: MZO2AØU.

Scandiano, 15/07/2021



Spett.le **LAMINAM S.p.A.**

Via Ghiarola Nuova, 258
41042 Fiorano Modenese
(MO)

Rapporto di prova n. 0350/2001

n.rif.int.: 30583
Richiesto in data: 26/05/2021

Descrizione del campione

"Manufatto costituito da piastrella di ceramica, fibra di vetro e resina epossidica 10x10 cm contrassegnato:
LAMINAM "

Campionamento

effettuato dal committente

Data di ricevimento materiale

01/06/2021

Data di esecuzione prove

inizio: 09/06/2021 fine: 07/07/2021

Specifica di prova/ descrizione metodo /procedura di prova

" Determinazione della resistenza dei colori alla luce - Norma DIN 51094 "

La riproduzione del presente rapporto di prova è autorizzata solo in forma di fotocopia completa.
Per ogni riproduzione parziale è necessaria la nostra autorizzazione.
Il presente rapporto di prova è costituito da 2 pagine (compresa questa).
I risultati riportati si riferiscono solo ai campioni esaminati.

Main Laboratory Sassuolo S.r.l. - Via Fjli Setti, 7 - 42019 Scandiano (RE) - tel. +39 0522 982603 - fax +39 0522 852834 - e-mail: info@mainlaboratorysassuolo.it
Reg. Impr. di RE - Cod. Fisc. e P.IVA 03164360368 - Cap. Soc. € 40.000,00 - R.E.A. di RE 269950

Rapporto di prova n. 0350/2001 Pag.2 di 2 Data 15/07/2021

LAMINAM S.p.A.

Campione

"Manufatto costituito da piastrella di ceramica, fibra di vetro e resina epossidica 10x10 cm contrassegnato: LAMINAM "

DETERMINAZIONE DELLA RESISTENZA DEI COLORI ALLA LUCE (Norma DIN 51094)

La prova consiste nell'esporre una parte della superficie dei campioni di prova all' azione di raggi ultravioletti per 28 giorni (la sorgente luminosa deve essere posta ad una distanza di 500 mm circa) in modo che i raggi colpiscano i campioni ad angolo retto.
Si considera che le piastrelle abbiano colori resistenti alla luce se, alla fine del periodo di esposizione, non si osserva nessuna variazione cromatica tra le parti esposte ai raggi ultravioletti e le parti non esposte.

Nota: il test è stato effettuato su campioni di tipologia diversa da quella per la quale la Norma è stata concepita.

Campione	Esame visivo
1	*nessuna variazione*
2	*nessuna variazione*
3	*nessuna variazione*
4	*nessuna variazione*
5	*nessuna variazione*

IL RESPONSABILE TECNICO
(I.Biorci)

Fire Resistant Certificate

07



ISTITUTO GIORDANO

ESTRATTO DI RAPPORTO DI CLASSIFICAZIONE N. 358672

ABRIDGED CLASSIFICATION REPORT No. 358672

Il presente documento si basa sul rapporto di classificazione n. 358187 emesso da Istituto Giordano in data 31 gennaio 2019

this document is based on classification report No. 358187 dated 31 January 2019 issued by Istituto Giordano

Cliente / *Customer*

LAMINAM S.p.A.

Via Ghiarola Nuova, 258 - 41042 FIORANO MODENESE (MO) - Italia

Oggetto / *Item**

lastra ceramica rinforzata con stuoia in fibra di vetro applicata sul retro con collante di spessore 3,4 mm e peso 8,25 kg/m² denominata "LAMINAM 3+"

ceramic sheet reinforced with fiberglass matting glued to the back, thickness 3,4 mm and weight 8,25 kg/m² named "LAMINAM 3+"

Attività / *Activity*

classificazione al fuoco dei prodotti e degli elementi da costruzione - Parte 1: Classificazione in base ai risultati delle prove di reazione al fuoco secondo la norma UNI EN 13501-1:2009

fire classification of construction products and building elements - Part 1: Classification using data from reaction to fire tests in accordance with standard UNI EN 13501-1:2009

Risultati / *Results*

in base ai risultati ottenuti dalle prove eseguite ai sensi delle norme UNI EN 13823:2014 e UNI EN ISO 1716:2010, all'oggetto in esame viene attribuita la

based on test result obtained in accordance with standards UNI EN 13823:2014 and UNI EN ISO 1716:2010, the item under examination may be rated as

Classificazione
Classification
A2 - s1, d0

(*) secondo le dichiarazioni del cliente.
according to that stated by the customer.

Bellaria-Igea Marina - Italia, 15 febbraio 2019
Bellario-Igea Marina - Italy, 15 February 2019

L'Amministratore Delegato
Chief Executive Officer
(Dott. Arch. Sara Lorenza Giordano)

Firmato digitalmente da SARA LORENZA GIORDANO

Commessa: / *Order:*
78185

Luogo dell'attività: / *Activity site:*
Istituto Giordano S.p.A. - Strada Erbosa Uno, 80 - 47043 Gatteo (FC) - Italia

Il presente documento è composto da n. 1 pagina (in formato bilingue (italiano e inglese), in caso di dubbio è valida la versione in lingua italiana) e non può essere riprodotto parzialmente, estrapolando parti di interesse a discrezione del cliente, con il rischio di favorire una interpretazione non corretta dei risultati, fatto salvo quanto definito a livello contrattuale.
I risultati si riferiscono solo all'oggetto in esame, così come ricevuto, e sono validi solo nelle condizioni in cui l'attività è stata effettuata.
Per quanto riguarda la descrizione dell'oggetto, i riferimenti normativi, le modalità, le apparecchiature, i risultati e quant'altro necessario all'identificazione dell'attività svolta si rimanda al rapporto di classificazione n. 358187 emesso da Istituto Giordano in data 31 gennaio 2019.
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The results relate only to the item examined, as received, and are valid only in the conditions in which the activity was carried out.
As regards the description of the item, apparatus, method, results and everything else necessary for the identification of the work carried out, please see classification report No. 358187 issued by Istituto Giordano on 31 January 2019.
The original of this document consists of an electronic document digitally signed pursuant to the applicable Italian Legislation.

Responsabile Tecnico: / *Chief Technician*
Dott. Ing. Giombattista Traina
Responsabile del Laboratorio di Reazione al Fuoco / *Head of Reaction to fire Laboratory:*
Dott. Ing. Giombattista Traina

Compilatore: / *Compiler:* Francesca Manduchi
Revisore: / *Reviewer:* Per. Ind. Andrea Golinucci

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PEC: ist-giordano@legalmail.it

Codice fiscale/Partita IVA: 00 549 540 409
Capitale sociale € 1.500.000 i.v.
REA, c/o C.C.I.A.A. (RN) 156766
Registro Imprese della Romagna - Forlì-Cesena e Rimini n. 00 549 540 409

Impact Load Test Certificate

09



EMINENT
INTERNATIONAL

FINAL REPORT

IMPACT TEST FOR

STONELAM SLAB

STONELAM SURFACES LLP

FR. No. : BE02/11/Aug/2023/MS 00
FR Date & Time : 11/Aug/2023 & 10:39PM
File Name : BE02-Stonelam Impact Test FR00

EMINENT INTERNATIONAL TESTING CENTRE PRIVATE LIMITED

Plot No 76/Part, EPIP, Industrial Park, Pashamailaram, Isnapur - 502307, Hyderabad, Telangana, India

Mail: info@eminent-itc.com **Website: www.eminent-itc.com**

EMINENT INTERNATIONAL TESTING CENTRE

DUBAI

Office No. F181, Hamid
Sultan Bin Muawaiz Building,
Ras Al Khor, Dubai, UAE.
M: +971 562850957.

INDIA

Plot No. 76/Part, EPIP, Industrial Park,
Pashamailaram, Isnapur – 502307,
Hyderabad, Telangana, India.
M: +91 7025000043.

ACCREDITATION

LABORATORY AND SITE TESTING

ISO / IEC 17025: General requirements for the competence of testing and calibrating laboratories by

NABL – National Accreditation Board for Testing & Calibration Laboratories

www.nabl-india.org

The testing standard(s) which is / are a subject of this document falls wholly or partly under the accreditation checked below

ISO / IEC 17025 – NABL	☐
It is not within our accreditation scope	☑

TC-11560

EMINENT
INTERNATIONAL

1. GENERAL

This is the final report for testing conducted by Eminent International Testing Centre to determine the performance impact adequacy of slab which was supplied and installed by Stonelam Surfaces LLP. The tests were performed in conformance with the extracted test procedures from *BE EN 12600* standard.

PROJECT DETAILS	
Project Name	
Client	
Consultant	
Architect	
Slab Contractor	Stonelam Surfaces LLP

TEST SPECIMEN & PERTINENT INFORMATION	
Manufacturer	Stonelam Surfaces LLP
Type	Slab
Operation Type	Fixed
Specimen Flat / Curve	Flat
Slab Thickness	3mm
Weather Seal Dimension	5mm
Weather Seal Type	789 Dow Corning (non staining)
Screws Dimension	6 x 15mm screws
Aluminium Profile	40 x 40mm

SPECIMEN LOCATION & DETAILS	
Test Location	EITC, Hyderabad, India

2. TEST PARAMETERS AND PERFORMANCE REQUIREMENTS

Test	Criteria	Reference
1. Impact Test		
Impactor	Dual tyre impactor, 50 ± 0.1 kg	Note: this was an ad hoc testing and test procedure and drop height extracted from BS EN 12600
Drop height	190mm, 450mm, 1200 mm	

Note: The testing criteria were taken from the project specification and approved by the client & consultant.

3. DESCRIPTION OR MODIFICATION OR ADJUSTMENTS

No any changes or modifications were done at the time of testing.

4. COMPLIANCE STATEMENT

The impact test for curtain wall was conducted as per extracted test procedure from BS EN 12600. The results of each test have been mentioned in this document and test results are valid only for date & time mentioned here.

5. SPECIFICATION COMPLIANCE

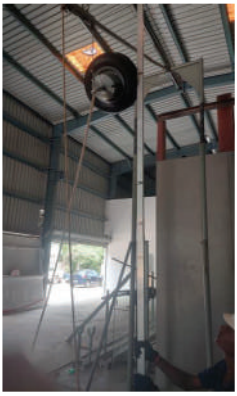
Testing was carried out under the direction of the Testing Engineer of EITC in compliance with the requirements of project specification and standards.

6. SUMMARY OF THE TEST

Tested locations were selected by client. Results for the tests conducted are below,

Sample	1	**Date**	08/08/2023 & 12:40PM
Impact Location	**Drop Height**	**Behaviour of specimen**	
Centre of slab	190	No crack or damages were observed	Photos has been attached below
	450	Crack was observed at middle. Specimen hold together and do not separate. Back support was bent.	
	1200	Cracks were observed at middle and both middle ends. Specimen hold together and do not separate. Back support was bent.	



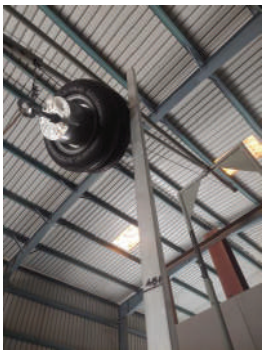


Sample	2	Date	08/08/2023 & 1:40PM
Impact Location	**Drop Height**	**Behaviour of specimen**	
Centre of slab	190	No crack or damages were observed	Photos has been attached below
	450	Crack was observed at middle. Specimen hold together and do not separate. Back support was bent.	
	1200	Cracks were observed at middle and both middle ends. Specimen hold together and do not separate. Back support was bent.	









Sample	3	Date	08/08/2023 & 1:55 PM
Impact Location	**Drop Height**	**Behaviour of specimen**	
Centre of slab	190	No crack or damages were observed	Photos has been attached below
	450	Crack was observed at middle. Specimen hold together and do not separate. Back support was bent.	
	1200	Cracks were observed at middle and both middle ends. Specimen hold together and do not separate. Back support was bent.	









Frost Resistant Certificate



16

MAIN LABORATORY SASSUOLO

LAB N° 1170 L

Testing, Experimentation and Quality Control Laboratory

Main Laboratory Sassuolo S.r.l. - Via F.lli Setti, 7 - 42019 Scandiano - Reggio Emilia (Italy) Cod.Fisc. e P.IVA 03164360368
Tel: 0522 982603 Fax: 0522 852834 E-mail: info@mainlaboratorysassuolo.it Web: www.mainlaboratorysassuolo.it

TEST REPORT n. 1411/2022/I

CERAMIC TILES

SASO ISO 10545-12:2007

DETERMINATION OF FROST RESISTANCE

Date of report:	03/31/2022
Customer:	**LAMINAM S.p.A.** Via Ghiarola Nuova, 258 41042 Fiorano Modenese (MO)
Requested on:	02/07/2022
Our ref.number:	32482
Execution place of tests:	Scandiano (RE)
Description of the sample:	"Unglazed ceramic tiles 500x500 mm marked: UNGLAZED CERAMIC TILES Bla UGL RECTIFIED"
Sampling:	carried out by the customer
Receipt date of samples:	03/08/2022
Execution date of tests:	start: 03/13/2022 end: 03/22/2022
Test specification:	SASO ISO 10545-12:2007 Ceramic tiles - Determination of frost resistance
Warnings:	*This test report may not be reproduced in part without our written approval.* *The results reported only refer to the samples tested, as received, and are only valid under the conditions in which the work was carried out.* *The information enclosed in inverted commas was provided by the customer and the laboratory accepts no liability for it.*

MAIN LABORATORY SASSUOLO

LAB N° 1170 L

Testing, Experimentation and Quality Control Laboratory

Main Laboratory Sassuolo S.r.l. - Via F.lli Setti, 7 - 42019 Scandiano - Reggio Emilia (Italy) Cod.Fisc. e P.IVA 03164360368
Tel: 0522 982603 Fax: 0522 852834 E-mail: info@mainlaboratorysassuolo.it Web: www.mainlaboratorysassuolo.it

TEST REPORT n. 1411/2022/I

CERAMIC TILES

SASO ISO 10545-12:2007

DETERMINATION OF FROST RESISTANCE

Principle:	after impregnation with water, submission of the tiles to a cycle between + 5°C and - 5°C, all sides of the tiles being exposed to freezing during a minimum of 100 freeze-thaw cycles.
Number of tested tiles:	10
Initial water absorption, E_1 (%):	0,0
Final water absorption, E_2 (%):	0,0
Number of damaged tiles after 100 cycles:	0
Defects before the test:	no visible defect.
Damage caused after the test:	no visible alteration.

MAIN LABORATORY SASSUOLO S.R.L. Centro di Ricerca Sperimentazione e Controllo Qualità

The Director
Giulia Gaido

End of the document

Requirements (Standard SASO ISO 13006:2018 - Table G.1)

Required

Note: the test has been carried out on 50x50 cm samples obtained by the Client by cutting slabs of dimensions 162x324 cm, 120x270 cm and 100x300 cm.

Water Absorption Certificate



19

MAIN LABORATORY SASSUOLO

LAB N° 1170 L

Testing, Experimentation and Quality Control Laboratory

Main Laboratory Sassuolo S.r.l. - Via F.lli Setti, 7 - 42019 Scandiano - Reggio Emilia (Italy) Cod.Fisc. e P.IVA 03164360368
Tel: 0522 982603 Fax: 0522 852834 E-mail: info@mainlaboratorysassuolo.it Web: www.mainlaboratorysassuolo.it

TEST REPORT n. 1109/2022/I

CERAMIC TILES

SASO ISO 10545-3:2018

DETERMINATION OF WATER ABSORPTION

Date of report:	03/21/2022
Customer:	**LAMINAM S.p.A.** Via Ghiarola Nuova, 258 41042 Fiorano Modenese (MO)
Requested on:	02/07/2022
Our ref.number:	32482
Execution place of tests:	Scandiano (RE)
Description of the sample:	"Unglazed ceramic tiles 500x500 mm marked: UNGLAZED CERAMIC TILES BIa UGL RECTIFIED"
Sampling:	carried out by the customer
Receipt date of samples:	03/08/2022
Execution date of tests:	start: 03/17/2022 end: 03/18/2022
Test specification:	SASO ISO 10545-3:2018 Ceramic tiles - Determination of water absorption
Warnings:	*This test report may not be reproduced in part without our written approval.* *The results reported only refer to the samples tested, as received, and are only valid under the conditions in which the work was carried out.* *The information enclosed in inverted commas was provided by the customer and the laboratory accepts no liability for it.*

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Tel: 0522 982603 Fax: 0522 852834 E-mail: info@mainlaboratorysassuolo.it Web: www.mainlaboratorysassuolo.it

TEST REPORT n. 1109/2022/I

CERAMIC TILES

SASO ISO 10545-3:2018

DETERMINATION OF WATER ABSORPTION

Principle: impregnation of dry tiles dried and calculation their water absorption Ev.

N. samples tested: 5

Sample size: before preparation: 50x50 cm after: 20x20 cm

Test results: 1) Water absorption E_v (%):

n. sample	E_v (%)
1	0,00
2	0,01
3	0,00
4	0,01
5	0,01

2) Water absorption mean value E_v (%): 0,01

MAIN LABORATORY SASSUOLO S.R.L. Centro di Ricerca Sperimentazione e Controllo Qualità

The Director
Giulia Gaido

End of the document

Requirements (Standard SASO ISO 13006:2018 - Table G.1)

$E_V \leq 0,5\%$

Individual maximum 0,6%

Note: the test has been carried out on 50x50 cm samples obtained by the Client by cutting slabs of dimensions 162x324 cm, 120x270 cm and 100x300 cm.

Wind Load Test Certificate



22

TEST REPORT NO: FITI/LAB/22/02/27

Test Report
For
Laboratory Performance Testing of Dry Cladding System.

: Project :
SFS1

: Testing For :
M/s. STONELAM SURFACES LLP.
D-176, Mansarovar Garden, New Delhi – 110015

: Testing Laboratory :
FAÇADE INDIA TESTING INC
H-23, Additional MIDC, Kudavali Village
Murbad, Maharashtra State 421401.

Mock up Testing carried out on
3rd February 2022.

Test report prepared on 4th February 2022.

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TEST REPORT

1.	PROJECT	: SFS1(STONELAM-SYSTEM TEST)
2.	TESTING LABORATORY	: Façade India Testing Inc. Plot No.H-23, Additional MIDC, Murbad, Thane District, Maharashtra 421401.
3.	TEST REPORT No.	: FITI/LAB/22/02/27
4.	TEST CONDUCTED FOR	: M/s. STONELAM SURFACES LLP D-176, Mansarovar Garden, New Delhi-110015.
5.	PROJECT ADDRESS	: NA
6.	TEST METHOD USED	: As per ASTM standards.
7.	METHOD STATEMENT	: FITI/MET/22/01/29.
8.	SPECIMEN RECEIVED	: 25th January 2022
9.	SPECIMEN CONDITION	: Good.
10.	TEST CONDUCTED ON	: 3rd February 2022.
11.	TEST REPORT ISSUED ON	: 4th February 2022.
12.	TEST RESULTS	: As per table below.

Façade India Testing Inc

TEST DATE	TEST	AS PER STANDARD	TEST PRESSURE	PERMISSIBLE LIMIT	MEASURED VALUE
03.02.2022	Structural Performance @ DWP	ASTM E330/ E330M-14(2021)	+2000 Pa	Deflection of Cladding Sheet & Structural members as per given criteria	Deflection of all members found within permissible limit
03.02.2022	Structural Performance @ DWP	ASTM E330/ E330M-14(2021)	-2000 Pa	Deflection of Cladding Sheet & Structural members as per given criteria	Deflection of all members found within permissible limit
03.02.2022	Structural Performance @150% DWP	ASTM E330/ E330M-14(2021)	+3000 Pa	Deformation of structural members as per given criteria. No failure of any component.	Deformation found within permissible limit. Also, no component failure observed.
03.02.2022	Structural Performance @150% DWP	ASTM E330/ E330M-14(2021)	-3000 Pa	Deformation of structural members as per given criteria. No failure of any component.	Deformation found within permissible limit. Also, no component failure observed.

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13.	COMPLIANCE STATEMENT	: Dry Cladding system specimen was tested as described in this document and method statement.

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FACADE INDIA TESTING INC MURBAD 421 401

V.S. Raghuraman
(Quality Manager)

Facade India Testing Inc

1.0 Introduction:

This Test Report is for full scale laboratory testing of Dry Cladding System for the project SFS1.

Full scale mockup testing was conducted as described herein, in accordance with the performance requirements of the project specifications and ASTM testing standards.

Testing was performed under an agreement with M/s. STONELAM SURFACES LLP and FAÇADE INDIA TESTING INC.Copies of Mockup details and project specifications are included in this document.

2.0 Project Details:

Project : SFS1

Customer : M/s. STONELAM SURFACES LLP.

3.0 Test Specimen:

Test specimen was full size Dry Cladding System of size 3000 mm Wide X 3000 mm Height.The mockup was installed by customer, on the structural test chamber, provided by the test laboratory. The test chamber was sealed at the perimeter of the curtain wall by sealant and was allowed to cure for 2-3 days. The test chamber was connected to test equipments prior to test. Test specimen drawing is enclosed in this document as annexure.

Mock-up system	: Dry Cladding System
Size	: Width 3 m x Height 3 m
Total area	: 9 m^2

4.0 Design Wind Pressure:

Project Design Wind Pressure is ±2.0 kPa(±2000 Pa).

4.1 Testing Standards:

The tests were carried out in accordance with following standard:

4.1.1 Structural Performance As per ASTM E330/E330M-14(2021).

Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference.

4.2 Testing Requirements:

a) Pre Load at 50% of + ve Design Wind Pressure.
b) Structural Performance test at +ve Design Wind Pressure= +2000 Pascal.
c) Structural Performance test at –ve Design Wind Pressure= -2000 Pascal.
d) Structural Performance test at 150% +ve Design Wind Pressure = +3000 Pascal.
e) Structural Performance test at 150% -ve Design Wind Pressure = -3000 Pascal.
f) Concentrated Load test at Panel Centre: 115 kg

g) Concentrated Load test at Panel Centre up to failure: 185 kg.

4.3 Testing Equipments:

a) Bi-Directional Vortex Blower & VFD.

b) 8" Hose Duct.

c) Test Chamber.

d) Hydraulic Jack with Power Pack.

4.4 Testing Measurement Equipments:

a) Digital Differential Pressure Transmitter.

b) U Type Manometer.

c) Hygrometer–Temperature & Relative Humidity Measurement.

d) Deflection Transducers.

e) IPC & SCADA Software.

f) Load Cell with indicator

5.0 Test Description:

5.1 Pre Load test at 50% of +ve Design Wind Pressure.

Test was conducted on 3rd February 2022. A pre load equivalent to 50% positive design load applied to the chamber and specimen held for 10 sec before returning back to zero. The purpose to stabilize the specimen with test chamber. It was decided to proceed with Structural Performance Test at Design Wind Pressure.

5.2 Structural Performance Test @ Design Wind Pressure:

Structural Performance Test @ DWP chamber -ve

Structural Performance Test at Design Wind Pressure Chamber Negative was performed on 3rd february 2022 after completion of Pre Load Test. The structural performance test was conducted as described in the method statement.The deflection transducers were placed as shown in the drawing diagram and initial values were set zero. The pressure was decreased upto -1000 Pa (50% of design wind pressure) and held for 10 seconds and the deflection values were recorded before bringing back to the zero. Residual deflection values were set zero. After 3 min, chamber pressure decreased upto -2000 Pa (100% of design wind pressure) in chamber and held for 10 seconds and the deflection values were recorded before bringing back to the zero. The permissible deflection Vs actual deflections were compared. The actual deflections were found within permissible limit. Residual deflections were observed zero. The test was recorded as Compliance. The test record sheets with deflection values are included in the Annexure. It was decided to proceed with Structural Performance Test at Design Wind Pressure Chamber Positive.

Structural Performance Test @DWP chamber +ve

Structural Performance Test at Design Wind Pressure Chamber Positive was performed on 3rd February 2022 immediately after completion of Structural Performance Test at Design Wind Pressure Chamber Negative. The structural performance test was conducted as described in the method statement. The deflection transducers were placed as shown in the drawing diagram and initial values were set zero. The pressure was increased up to +1000 Pa (50% of design wind pressure) and held for 10 seconds and the deflection values were recorded before bringing back to the zero. Residual deflection values were set zero. After 3 min, chamber pressure increased up to +2000 Pa (100% of design wind pressure) in chamber and held for 10 seconds and the deflection values were recorded before bringing back to the zero. The permissible deflection Vs actual deflections were compared. The actual deflections were found within permissible limit. Residual deflections were observed zero. The test was recorded as Compliance. The test record sheets with deflection values are included in the Annexure. It was decided to proceed with Structural Performance Test @ 150% of DWP.

5.3 Structural Performance Test:

At 150% of Design Wind Pressure (chamber–ve)

Structural Performance Test Chamber Negative at 150% Design Wind Pressure was conducted on 3rd February 2022. Prior to the test initial values of deflection transducers were set zero. The chamber pressure was decreased up to -1500 Pa (75% of DWP) and held for 10 seconds, before bringing back to zero Pascal. Subsequently, chamber pressure was decreased up to -3000 Pa (150% of DWP) and held for 10 seconds before brining back to the zero. After waiting for five minute the residual deformation were recorded with the help of transducers. The actual deformation was compared and found within permissible limits. Also, no component failures were observed. Copy of test record sheet included in the Annexure. It was decided to proceed with Structural Performance Test Chamber Positive at 150% Design Wind Pressure.

At 150% of Design Wind Pressure (chamber+ve)

Structural Performance Test Chamber Positive at 150% Design Wind Pressure was conducted on 3rd February 2022. Prior to the test initial values of deflection transducers were set zero. The chamber pressure was increased up to +1500 Pa (75% of DWP) and held for 10 seconds, before bringing back to zero. Subsequently, chamber pressure was increased up to +3000 Pa (150% of DWP) and held for 10 seconds before bringing back to the zero. After waiting for five minute the residual deformation were recorded with the help of transducers. The actual deformation was compared and found within permissible limits. Also, no component failures were observed. The test was recorded as Compliance. Copy of test record sheet included in the Annexure. It was decided to proceed with Concentrated Load Test at Panel Centre.

5.4 Concentrated Load Test at Cladding Panel Centre :115 kg

Concentrated Load Test at Panel Centre was performed on 3rd February 2022 immediately after completion of Structural Performance Test Chamber Positive at

150% Design Wind Pressure. The test was conducted for 115 kg concentrated load applied horizontally in push direction at one panel. The concentrated load was applied horizontally in push direction from outside at 300 mm x 300 mm surface area with the help of load cell and hydraulic jack. Maximum load was held for 60 seconds and deflection value of panel was recorded from inside. No failure of any component was observed. The test was recorded as Compliance. Copy of test record sheet included in the Annexure.

5.5 Concentrated Load Test at Panel Centre up to failure: 185 kg

Concentrated Load Test at Cladding Panel Centre up to failure was performed on 3rd February 2022 immediately after completion of concentrated load test @115 kg. The test was conducted to check maximum reserved strength of sheet. The concentrated load was applied horizontally in push direction from outside at 200 mm x 100 mm surface area. The load was applied gradually with the help of hydraulic jack. At 185 kg, sheet breakage was observed from inside and outside. The test was conducted for study purpose.

6.0 Test Witnesses:

The final testing was witnessed in whole by the following:

1	Mr. Sandeep Bagade	Customer	M/s. STONELAM SURFACES LLP
2	Mr. Sangh Pradeep	Customer	M/s. STONELAM SURFACES LLP
3	Mr. Chandra Prakash Vishwakarma	Customer	M/s. STONELAM SURFACES LLP

7.0 Cladding Wall System Test Results Summary:

Dry Cladding System was tested as described in this document and method statement and test results reported under clause 5.

This report is issued on behalf of Façade India Testing Inc.

For Façade India Testing Inc.

FACADE INDIA TETING INC MURBAD 421 401

V.S. Raghuraman
(Quality Manager)

PROJECT : SYSTEM TEST (STONELAM SURFACES LLP).

SYSTEM : DRY CLADDING SYSTEM.

MOCK UP PERFORMANCE TEST (PRE TEST)

FITI/LAB/22/02/27

DATE:01.02.2022

Sr. No.	Witnessing Person	Name of Organization	Sign
01.	Sangh Pradeep	Stonelam Surfaces LLP	
02.	Chandra Prakash Vishwakarma	Stonelam Surfaces LLP	
03.			
04.			
05.			
06.			
07.			
08.			
09.			
10.			
11.			
12.			
13.			

FACT INDIA TESTING INC HYDERABAD 421

PROJECT : SYSTEM TEST (STONELAM SURFACES LLP).

SYSTEM : DRY CLADDING SYSTEM.

MOCK UP PERFORMANCE TEST (FINAL TEST)

FITI/LAB/22/02/27

DATE:03.02.2022

Sr. No.	Witnessing Person	Name of Organization	Sign
01.	Sandeep Bagade	Stonelam Surfaces LLP	
02.	Singh Pradeep	Stonlam Surfaces LLP	
03.	Chandra Prakash Vishwakarma	Stonelam Surfaces LLP	
04.	Rajmani Vishwakarma	PAV INFRASTRUCTURE PVT LTD	03/02/2022
05.			
06.			
07.			
08.			
09.			
10.			
11.			
12.			
13.			

FACADE INDIA TESTING INC MURBAD 421 401

FAÇADE INDIA TESTING INC

PERFORMANCE TEST REPORT SUMMARY

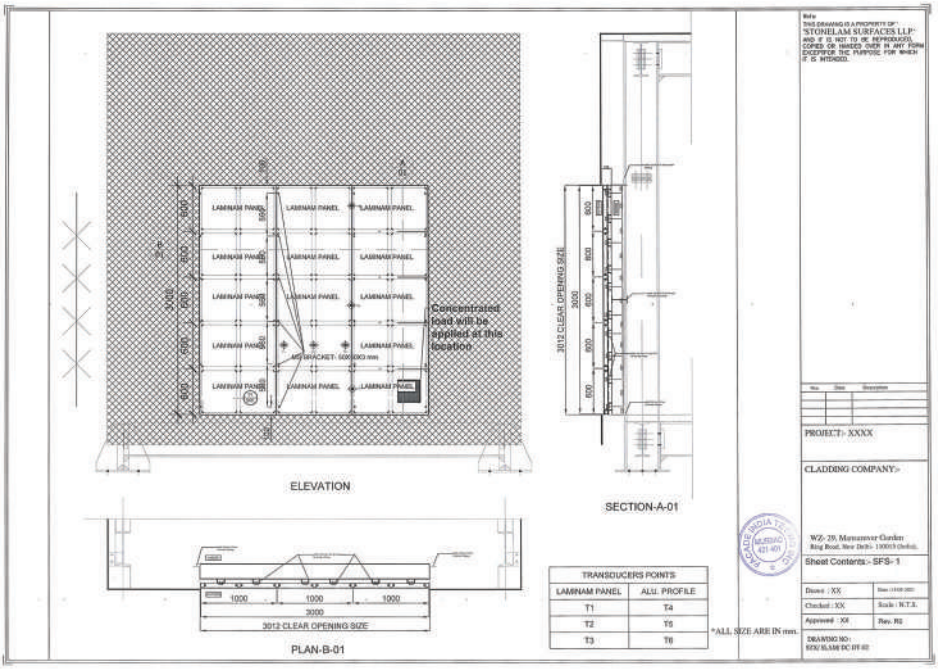
PROJECT: SFS1 [STONELAM-SYSTEM TEST]
SYSTEM: DRY CLADDING SYSTEM.

Test Date: 03.02.2022

SR. NO.	TEST DATE	TEST REPORT PAGE #	NAME OF TEST	AS PER STANDARD	TEST PRESSURE	TEST DURATION	PARAMETER TO TEST	PERMISSIBLE LIMIT	MEASURED VALUE	INFERENCE	REMARKS
1	03.02.2022	1	Structural Performance @ SWP	ASTM E330/E330M-14(2021)	+2000 Pa	10 Min.	Deflection of Aluminium members including cladding panel at +2000 Pa.	———		Compliance	Deflection values of all members found within permissible limit. No Failure of any component.
							Laminam panel-1	2.5 mm	1.80 mm		
							Laminam panel-2	2.5 mm	1.96 mm		
							Aluminium profile	16.11 mm	7.42 mm		
2	03.02.2022	2	Structural Performance @ SWP	ASTM E330/E330M-14(2021)	-2000 Pa	10 Min.	Deflection of Aluminium members including cladding panel at -2000 Pa.	———		Compliance	Deflection values of all members found within permissible limit.No Failure of any component.
							Laminam panel-1	2.5 mm	1.68 mm		
							Laminam panel-2	2.5 mm	1.88 mm		
							Aluminium profile	16.11 mm	6.85 mm		
3	03.02.2022	3	Structural Performance @ 150% SWP	ASTM E330/E330M-14(2021)	+1000 Pa	10 Min.	Permanent deformation or cladding panel failure.	Allowable permanent deformation value		Compliance	Deformation values of all members found within permissible limit. No failure of any component.
							Laminam panel-1	0.50 mm	0.15 mm		
							Laminam panel-2	0.50 mm	0.21 mm		
							Aluminium profile	2.82 mm	2.16 mm		
4	03.02.2022	5	Structural Performance @ 150% SWP	ASTM E330/E330M-14(2021)	-3000 Pa	10 Min.	Permanent deformation or cladding panel failure.	Allowable permanent deformation value		Compliance	Deformation values of all members found within permissible limit. No failure of any component.
							Laminam panel-1	0.50 mm	0.14 mm		
							Laminam panel-2	0.50 mm	0.21 mm		
							Aluminium profile	2.82 mm	1.40 mm		
5	03.02.2022	4	Concentrated Push Load Test	As per customer specification	115 KG	10 Min	To check strength of cladding panel against concentrated load @ 115 KG	No failure of any component	No failure of any component	Compliance	Load applied on 300mm X 300mm area . No failure of any component observed
6	03.02.2022	4	Concentrated Push Load Test up to failure	As per customer specification	185 KG	10 Min	To check maximun strength of cladding panel against concentrated load	Study Purpose	Sheet breakage observed at 185 kg	NA	Load applied on 300mmX100mm area. Laminam cladding sheet broken has reserved flexural strength upto 185 kg.

TESTING ENGG.

AUTHORISED BY.



STRUCTURAL PERFORMANCE TEST AT 100 % DESIGN WIND PRESSURE

ASTM E 330/E330M-14(2021) (CHAMBER-VE)

FITI/LAB/22/02/27

TEST PRESSURE DIFFERENTIAL = 2000 Pa

DATE: 03/02/2022 14:27

PROJECT NAME: SFS1(STONELAM-SYSTEM TEST)

TEST CONDITION AT SITE:	OUT SIDE	IN SIDE
AMBIENT TEMPERATURE:	31.5 °C	29.5 °C
RELATIVE HUMIDITY:	60.0 %RH	75.0 %RH

Sl. No.	ITEM	CHANNEL NO.'S.	DIFF. PRESSURE (Pa)	DEFLECT @ A (mm)	DEFLECT @ B(mm)	DEFLECT @ C (mm)	ACTUAL CENTRAL DEFLECTION =B-(A+C)/2(mm)	SPAN (L) mm	ALLOWABLE DEFLECTION (mm)	CONCLUSION
1	LAMINAM PANEL-1	T1=Ch 8 T2=Ch 6 T3=Ch 1	50%= 1000 pa	T1= 0.60	T2= 2.66	T3= 3.05	0.84	500	2.50	C
			100%= 2000 pa	T1= 1.55	T2= 6.72	T3= 8.10	1.90			
2	LAMINAM PANEL-2	T4=Ch 19 T5=Ch 4 T6=Ch 12	50%= 1000 pa	T4= 1.60	T5= 3.41	T6= 3.47	0.88	500	2.50	C
			100%= 2000 pa	T4= 3.95	T5= 8.38	T6= 8.90	1.96			
3	ALU.PROFILE	T7=Ch 7 T8=Ch 1 T9=Ch 13	50%= 1000 pa	T7= 0.10	T8= 3.05	T9= 0.39	2.81	2820	16.11	C
			100%= 2000 pa	T7= 0.38	T8= 8.10	T9= 0.99	7.42			
4	NA	T10=Ch 44 T11=Ch 44 T12=Ch 44	50%= 1000 pa	T10= 0.00	T11= 0.00	T12= 0.00	0.00	0	0.00	NA
			100%= 2000 pa	T10= 0.00	T11= 0.00	T12= 0.00	0.00			
5	NA	T13=Ch 44 T14=Ch 44 T15=Ch 44	50%= 1000 pa	T13= 0.00	T14= 0.00	T15= 0.00	0.00	0	0.00	NA
			100%= 2000 pa	T13= 0.00	T14= 0.00	T15= 0.00	0.00			

NOTE:

TEST RESULT: **COMPLIANCE.**

Test Engg

Authorised

STRUCTURAL PERFORMANCE TEST AT 100 % DESIGN WIND PRESSURE

ASTM E 330/E330M-14(2021) (CHAMBER+VE)

FITI/LAB/22/02/27

TEST PRESSURE DIFFERENTIAL = -2000 Pa

DATE: 03/02/2022 14:34

PROJECT NAME: SFS1(STONELAM-SYSTEM TEST)

TEST CONDITION AT SITE:	OUT SIDE	IN SIDE
AMBIENT TEMPERATURE:	31.5 °C	29.5 °C
RELATIVE HUMIDITY:	60.0 %RH	75.0 %RH

Sl. No.	ITEM	CHANNEL NO.'S.	DIFF. PRESSURE (Pa)	DEFLECT @ A (mm)	DEFLECT @ B(mm)	DEFLECT @ C (mm)	ACTUAL CENTRAL DEFLECTION =B-(A+C)/2(mm)	SPAN (L) mm	ALLOWABLE DEFLECTION (mm)	CONCLUSION
1	LAMINAM PANEL-1	T1=Ch 8 T2=Ch 6 T3=Ch 1	50%= 1000 pa	T1= 0.70	T2= 2.82	T3= 3.28	0.83	500	2.50	C
			100%= 2000 pa	T1= 1.65	T2= 6.32	T3= 7.63	1.68			
2	LAMINAM PANEL-2	T4=Ch 19 T5=Ch 4 T6=Ch 12	50%= 1000 pa	T4= 1.77	T5= 3.57	T6= 3.65	0.86	500	2.50	C
			100%= 2000 pa	T4= 4.36	T5= 7.74	T6= 7.76	1.68			
3	ALU.PROFILE	T7=Ch 7 T8=Ch 1 T9=Ch 13	50%= 1000 pa	T7= 0.16	T8= 3.28	T9= 0.43	2.99	2820	16.11	C
			100%= 2000 pa	T7= 0.44	T8= 7.63	T9= 1.12	6.85			
4	NA	T10=Ch 44 T11=Ch 44 T12=Ch 44	50%= 1000 pa	T10= 0.00	T11= 0.00	T12= 0.00	0.00	0	0.00	NA
			100%= 2000 pa	T10= 0.00	T11= 0.00	T12= 0.00	0.00			
5	NA	T13=Ch 44 T14=Ch 44 T15=Ch 44	50%= 1000 pa	T13= 0.00	T14= 0.00	T15= 0.00	0.00	0	0.00	NA
			100%= 2000 pa	T13= 0.00	T14= 0.00	T15= 0.00	0.00			

NOTE:

TEST RESULT: **COMPLIANCE.**

Test Engg

Authorised

STRUCTURAL PERFORMANCE TEST AT 150 % DESIGN WIND PRESSURE
ASTM E 330/E330M-14 (2021)

FITI/LAB/22/02/27

TEST PRESSURE DIFFERENTIAL = 3000 Pa
TEST PRESSURE DIFFERENTIAL = -3000 Pa

DATE : 03/02/2022 15:15
PROJECT NAME : SFS1(STONELAM-SYSTEM TEST)
TEST CONDITION AT SITE : OUT SIDE | IN SIDE
AMBIENT TEMPERATURE : 31.5 °C | 30.0 °C
RELATIVE HUMIDITY : 72.0 %RH | 80.0 %RH

OBS. NO.	ELEMENT	PERMANENT DEFORMATION AT 3000 pa	PERMANENT DEFORMATION AT -3000 pa	SPAN mm	ALLOWABLE DEFORMATION mm	ANY ELEMENT FAILURE	CONCLUSION	
							AT 3000 pa	AT -3000 pa
1.	LAMINAM PANEL-1	0.15 mm	0.14 mm	500 mm	0.500	NO FAILURE	C	C
2.	LAMINAM PANEL-2	0.21 mm	0.21 mm	500 mm	0.500	NO FAILURE	C	C
3.	ALU.PROFILE	2.16 mm	1.40 mm	2820 mm	2.820	NO FAILURE	C	C
4.	NA	0.00 mm	0.00 mm	0 mm	0.000	NA	NA	NA
5.	NA	0.00 mm	0.00 mm	0 mm	0.000	NA	NA	NA
6.	NA	0.00 mm	0.00 mm	0 mm	0.000	NA	NA	NA

NOTE:

TEST RESULT : COMPLIANCE.

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Authorised

FITI/LAB/22/02/27

CONCENTRATED LOAD TEST
HORIZONTAL PUSH LOAD = 115 Kg

TEST TO FAILURE = 185Kg.

Date : 03/02/2022 15.40 PM

PROJECT NAME : SFS1(STONELAM SYSTEM TEST)

INTEST CONDITION AT SITE :

AMBIENT TEMPERATURE = 31.6 °C

RELATIVE HUMIDITY = 72.4 % RH

OBSERVATION NOS.	LOAD(Kg)	OBSERVED DEFLECTION AT PANEL CENTRE (mm)	ALLOWABLE DEFLECTION (mm)	CONCLUSION
1.	115 Kg	3.5 mm	NA	COMPLIANCE

NOTE: No Failure Of Any Component Observed.

2. TEST TO FAILURE @ 185 Kg:
OBSERVATION: A CONCENTRATED LOAD APPLIED OVER ONE LOCATION IN PUSH DIRECTION. LOAD ACHIEVED UPTO 185 KG BEFORE SHEET BREAKAGE. HENCE TEST CONCLUDED FOR MAXIMUM STRENGTH OF SHEET IS 185KG.

TEST RESULT: COMPLIANCE

Test Engg

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FITI/LAB/22/02/27



STRUCTURAL PERFORMANCE TEST @ 100% DWP = ±2000 Pa (CHAMBER±VE)

FITI/LAB/22/02/27



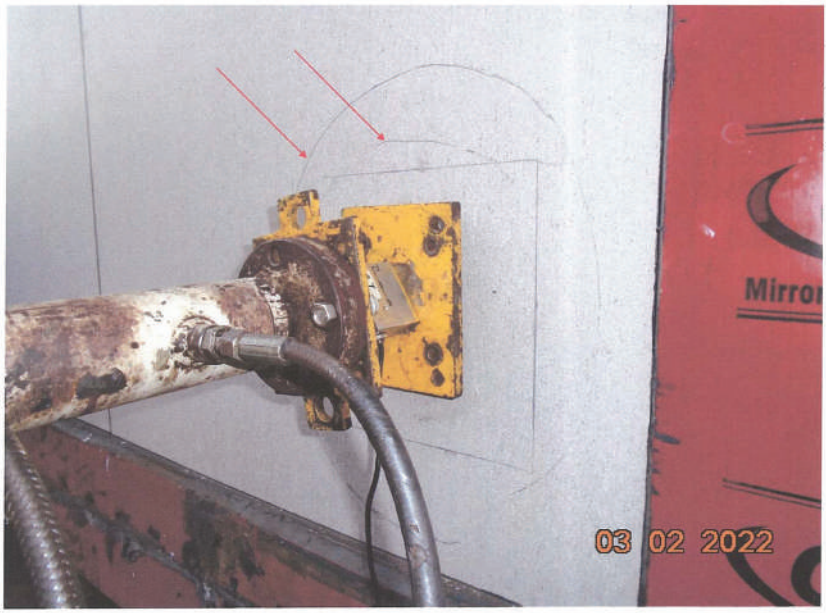
STRUCTURAL PERFORMANCE TEST @ 150% DWP = ±3000 Pa (CHAMBER±VE)

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CONCENTRATED LOAD TEST @ 115 KG

FIT/LAB/22/02/27



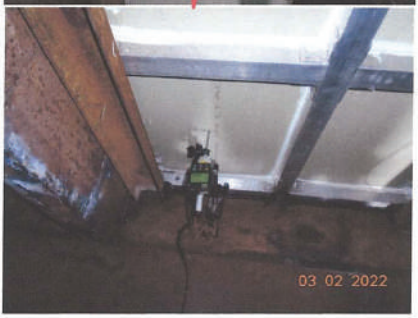
FACADE INDIA TESTING INC MURBAD 421 401

TEST TO FAILURE-CONCENTRATED LOAD TEST: 185 KG. BREAKAGE MARKS INDICATED BY ARROWS.

FIT/LAB/22/02/27







OBSERVATIONS TAKEN FROM INSIDE DURING CONCENTRATED LOAD TEST @ 115 KG AND 185 KG. NO BREKAGE WAS OBSERVED AT 115 KG. HOWEVER, CLADDING PANEL BRAEKAGE WAS OBSERVED AT 185 KG.

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FITI/LAB/22/02/27

CONTROLLED DISASSEMBLY REPORT
FOR PROJECT
SFS1.

Controlled Disassembly Date : 5th February 2022.

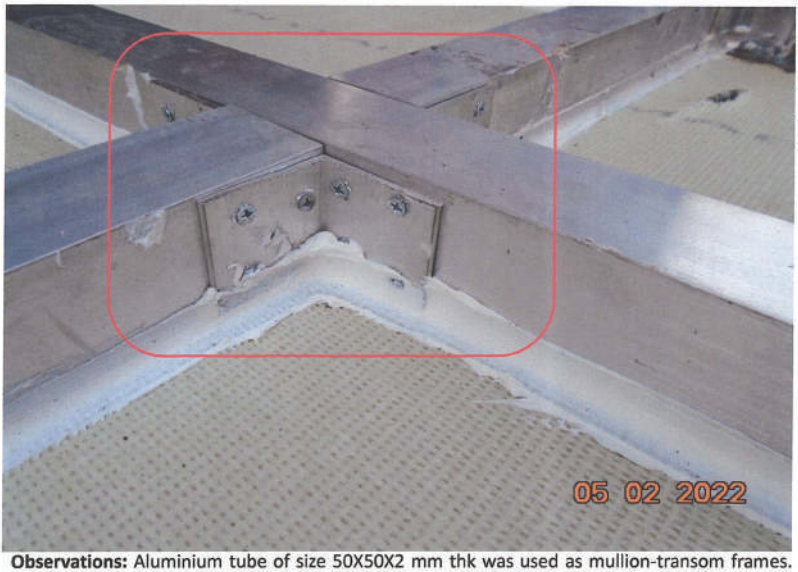
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FINAL TEST TEAM PHOTOGRAPH



FITI/LAB/22/02/27

Observations: Aluminium tube of size 50X50X2 mm thk was used as mullion-transom frames. Transoms were connected with Mullion by means of using 50X50X5mm thk Aluminium L-angle and CSK screws of size 6x16. Frames and cladding panel junction was sealed with sealant. No damage observed at any joint.



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Observations : Aluminum tube frame fixed with test rig by M8 X 75 nut-bolt.



Observation: Cladding panels were fixed with main frames with help of bonding tape sealant.



Observation: Sealant and bonding tape used in PMU installation.

Adhesion Test Certificate

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24 March 2021
Document ID: 539732

STONELAM® SURFACES LLP
LLPIN - AAO - 9695, WZ-29, Mansarovar Garden, Ring Road, New Delhi-110015, India.

Project Name: 21-003022 - General Testing (Adhesion testing of Stonelam Laminam Porcelain Slab)
Project Address: LLPIN - AAO - 9695, WZ-29, Mansarovar Garden, Ring Road, New Delhi-110015, India.

Ref.: ADHESION Test Report for Stonelam Laminam Porcelain Slab

The DOW construction laboratory has completed the adhesion testing in support of your above referenced project.

The adhesion tests are performed in accordance to the adapted ASTM C794. A summary of this laboratory testing / evaluation and application recommendations is given below.

1. Sample Nr 214948	**Substrate Details**	
Substrate Type	Slab	
Substrate Description	Stonelam Laminam Porcelain Slab	
Surface to Test	Back	
Cleaner	DOWSIL™ R-40 Universal Cleaner	
Sealant	**Primer**	**Application Recommendation**
DOWSIL™ 896 PanelFix for Bonding	DOWSIL™ Construction Primer P	Approved for Panel Bonding Applications

2. Sample Nr 214949	**Substrate Details**	
Substrate Type	Aluminum - Mill Finished	
Substrate Description	Mill Finished Aluminium Slim Clid Section	
Surface to Test	Mill Finish Surface	
Cleaner	DOWSIL™ R-40 Universal Cleaner	
Sealant	**Primer**	**Application Recommendation**
DOWSIL™ 896 PanelFix for Bonding	DOWSIL™ 1200 OS Primer	Approved for Panel Bonding Applications

Surfaces that are cleaned and primed today must be sealed today; otherwise recleaning and repriming are required before application.

Application recommendations are based on Dow laboratory testing, evaluation and technical specifications.

If sealant glazing will commence more than 12 months after this report is issued, please submit additional samples for re-test.

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silicones by DOW

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24 March 2021
Document ID: 539732

It is recommended that regular jobsite adhesion testing be performed throughout the project to confirm that good adhesion is obtained consistently on actual job site substrates. Information and literature on field adhesion testing and Dow application procedures may be obtained by contacting your Dow distributor or customer service representative.

The supplier of each structurally glazed substrate must be contacted to approve the use of its material in structural applications. Dow verifies only the adhesive characteristics and compatibility of the silicone sealant to those materials through laboratory testing and Dow assumes no responsibility or adequacy of those substrates.

Sincerely,

Vishakha Shende
Laboratory Technician

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LAMINAM
SUPERIOR NATURAL SURFACES

Fiorano Modenese (MO), 03/03/2022

Laminam S.p.A.
Via Ghiarola Nuova 258,
Fiorano Modenese (MO)
Italia

OGGETTO: LAMINAM – Adhesion performance of the fiberglass reinforcement

Laminam SpA manufactures multilayered slabs, composed by a ceramic slab and a glassfiber:

- Laminam 3+: Porcelain ceramic slab (nominal thickness 3mm) with reinforcement glassfiber applied onto the backside;
- Laminam 5+: Porcelain ceramic slab (nominal thickness 5.6mm) with reinforcement glassfiber applied onto the backside.

Laminam SpA applies the glassfiber as a reinforcement onto the back side of the ceramic slab with a controlled and automatic process, which allows to obtain high quality standards, verified by internal control procedure, and repeatability and consistency of product performance (process conforming to ISO 9001).

The evaluation of the final adhesion performances of the applied glassfiber and the maintenance of this performance were verified before and after several conditioning cycles.
The conditionings were selected as extreme conditions, simulating the worse conditions which Laminam products may undergo during the life of use of the final application, especially outdoor.

Conditionings	
28 d at 23°C (73,4 °F)	Standard condition
20 d at 50°C (122 °F)	Exposure to high temperature
41 d at 50°C (122°F)	
14 d at 70°C (158 °F)	
1 d at 80°C (176 °F)	
20 d in water	Immersion in water
21 d in water + 25 cycles -15/+15°C (5/+59 °F)	Immersion in water + Freeze/thaw
1 d at -20°C (-4°F)	Exposure to low temperatures
1 d at -40°C (-40°F)	
7 d in 3%NaCl solution	Saline environment
7 d in pH=3 solution	Immersion in acid solution
7 d in pH=13 solution	Immersion in basic solution

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LAMINAM
SUPERIOR NATURAL SURFACES

The test report confirms the good adhesion between the components making up the products mentioned:

- the adhesion resistance is above 2 N/mm2, considering the average value of the complete testing campaign;
- the adhesion resistance does not have significant decrease after each conditioning cycle compared to the standard conditions.

We take as a comparison 1 N/mm2, which is the pull-out resistance of C2-class cementitious adhesives, the pull-out resistance of suitable plaster for exterior cladding and the average pull-out resistance of structural adhesives, indicated in the principle international standards.

The complete test campaign was followed and reported by Istituto Giordano SpA (test reports n° 323607, 323622, 323623 and 323624), as an external Laboratory.

Sure to have communicated in the best technical information, we remain at your disposal for any further information.

Best regards

Laminam SpA
Project Engineering
fp/MB

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Food Grade Test Certificate



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NSF International

789 N. Dixboro Road, Ann Arbor, MI 48105 USA

RECOGNIZES

LAMINAM S.p.A.

Facility: Borgo Val Di Taro, Italy

AS COMPLYING WITH NSF/ANSI 51 AND ALL APPLICABLE REQUIREMENTS.
PRODUCTS APPEARING IN THE NSF OFFICIAL LISTING ARE
AUTHORIZED TO BEAR THE NSF MARK.

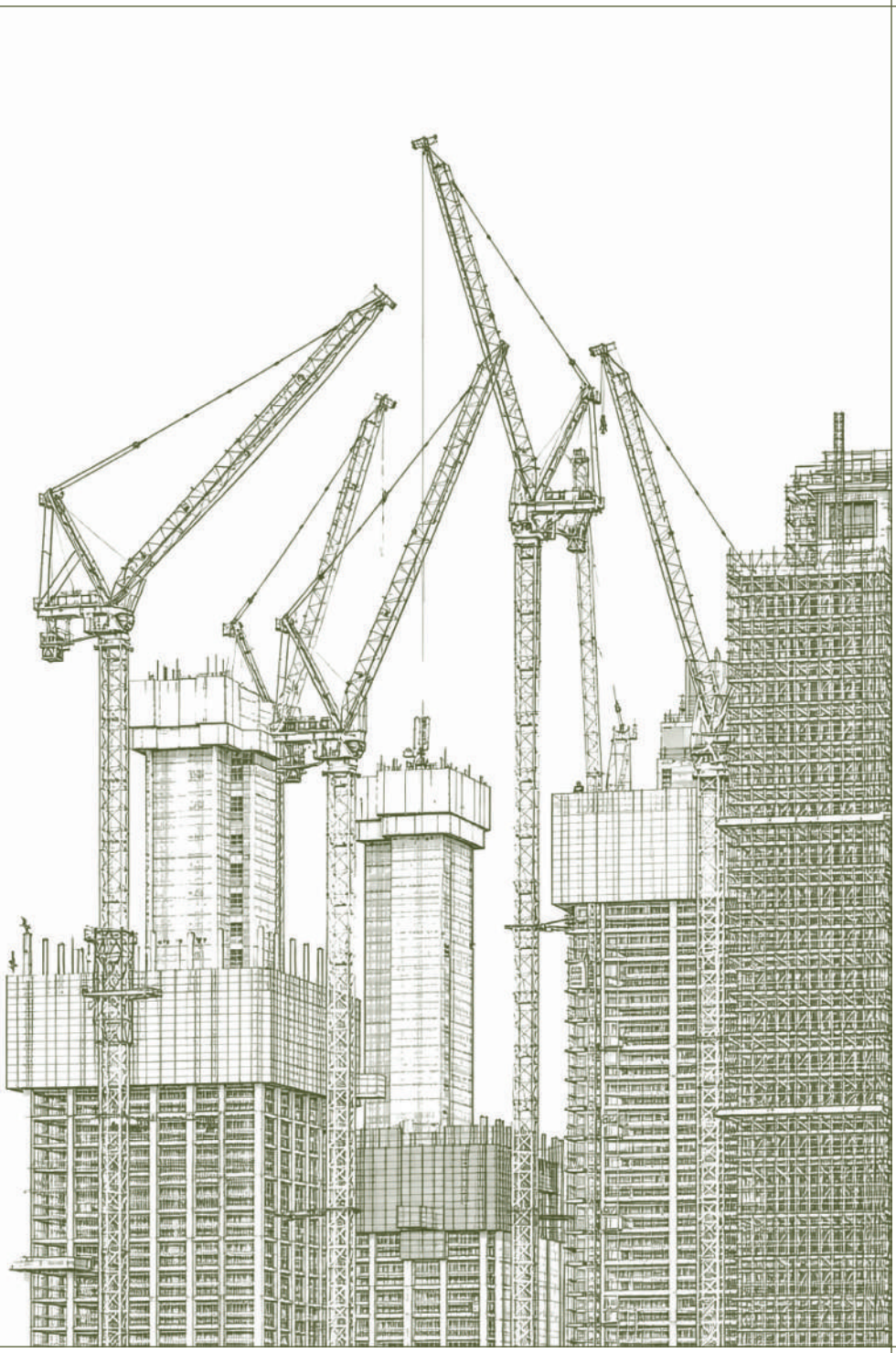
Certification Program Accredited by the American National Standards Institute

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May 22, 2018
Certificate# C0379503 - 01

Sarah Krol
Global Managing Director, Food Safety Product Certification



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